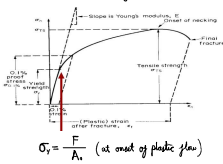


Loading a ductile metal in tension, it plastically deforms & curve becomes non-linear

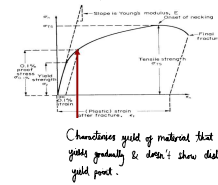
Yield Strength (σ_y)



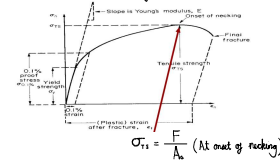
Permanent Set



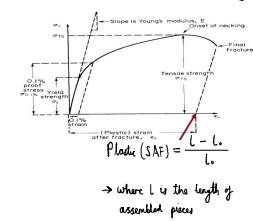
Proof Stress ($\sigma_{0.2}$)



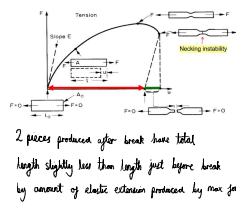
Tensile Strength (σ_t)



Strain after fracture / Tensile Ductility



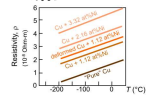
Permanent extension



Electrical Properties

Electrical resistivity

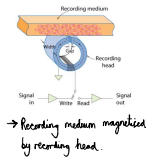
Example:



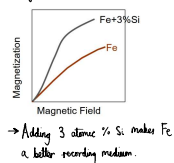
Increase resistivity of Cu by:
→ Adding impurity atoms
→ Degrading the Cu.

Magnetic Properties

Magnetic Storage



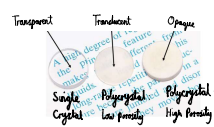
Magnetic Permeability Vs Composition



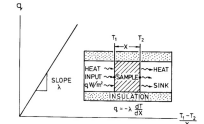
Optical Properties

Transmittance

→ Aluminium Oxide optical properties depend on material structure (shown below)



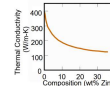
Thermal conductivity (λ)



→ Rate at which heat is conducted through solid at steady state.

$$q = -\lambda \frac{dT}{dx} = \lambda \frac{T_1 - T_2}{x}$$

Examples:

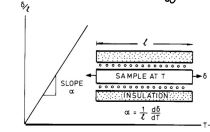


→ Thermal conductivity of Copper decreases as Zn is added.



→ Silicon film insulation offers low heat conduction

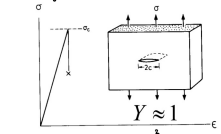
Thermal Expansion Coefficient



Thermal strain per degree = α (K^{-1})

→ If thermally isotropic, volume expansion per degree $\approx 3\alpha$

Toughness (G_c) / Fracture toughness (K_{IC})



$$G_c (KJ/m^2) = \frac{K_{IC}}{E(1+\nu)}$$

→ Energy to create new fracture surface.

$$K_{IC} (MPa m^{1/2} \text{ or } MNm^{3/2}) = Y \sigma_c \sqrt{\pi c}$$

→ Both measure resistance to propagation of crack

→ Measured by loading sample with deliberately introduced crack of 2c.

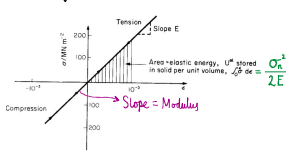
→ Tensile stress σ_c recorded when crack propagates

Density, ρ (Kg/m^3 or Mg/m^3 or $Kg/litre$)

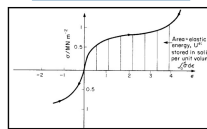
→ Important for reducing CO_2 emissions

Linear elastic behaviour

- Exhibited by most solids at low strains (0.001-0.1%) and obey Hooke's Law.
- Tension & Compression give same value
- All energy recovered upon unloading



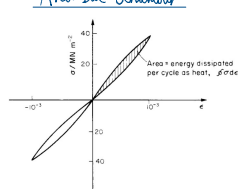
Non-linear Elastic



→ Rubbers exhibit this behaviour over large extensions (strains up to 500% are possible)

→ Most energy is recovered upon unloading

Anelastic Behaviour



→ Still elastic, so no permanent deflection.

→ Loading and unloading curves not the same, ∴ energy dissipated per cycle

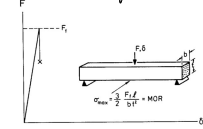
→ Sometimes desirable: damping and noise (rubber)

→ Damping is bad for bells (Spring steel, bronze, glass)

Different Moduli:

- Young's Modulus, E (GPa): describes tension or compression
 - $E = \frac{3G}{1 + \frac{2}{3}\nu}$
- Shear Modulus, G (GPa): describes shear loading
 - $G = \frac{E}{2(1+\nu)}$
- Bulk Modulus, K (GPa): describes hydrostatic behaviour
 - $K = \frac{E}{3(1-2\nu)}$
- Poisson's ratio, $\nu = -\frac{\epsilon_{\text{trans}}}{\epsilon_{\text{long}}}$
- Moduli measured accurately by exciting natural vibrations of beam or wire, or measuring velocity of waves in material

Modulus of Rupture



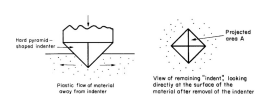
→ Is the surface stress in beam known at instant of failure

→ Used when it is difficult to grip sample

• For Ceramics: $1.3 \times \sigma_{TS}$, as volume subjected to max stress is small, hence probability of finding flaw is reduced

→ In simple tension all flaws see max stress.

Hardness Test



→ Pointed diamond or steel ball pressed into surface of examined material.

→ The further into the material it sinks, the softer the material; lower yield strength

$$H = \frac{F}{A}$$